

Lecture Summary and DNA Structure

Zachary Tullier

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Q1.

This lecture explains how radiation interacts with matter, how those interactions affect biological molecules and cells, and how the same principles are applied in medicine, environmental science, and space travel.

1 Radiation and Its Sources

Radiation spans a broad electromagnetic spectrum. Low-frequency radio waves, microwaves, infrared radiation, and visible light generally do not carry enough energy to ionize matter. X-rays and gamma rays have much shorter wavelengths and greater photon energies, allowing them to remove electrons from atoms and molecules.

Radiation comes from both natural and artificial sources:

- The Sun produces visible light, ultraviolet radiation, solar neutrinos, gamma rays, protons, and alpha particles.
- High-energy cosmic rays originate outside Earth and produce showers of secondary particles when they collide with atmospheric nuclei.
- Radioactive decay, nuclear reactions, particle accelerators, medical equipment, and nuclear technology provide terrestrial or artificial sources.

2 Ionizing Versus Non-Ionizing Radiation

Country List			
Possible bio-logical effects	Category	Examples	Primary inter-action
Non-ionizing	Radio waves, microwaves, infrared, visible light, and most ultraviolet radiation	Excites molecules or produces heat without normally removing electrons	Thermal effects, photochemical reactions, and some forms of DNA damage
Ionizing	High-energy ultraviolet radiation, X-rays, gamma rays, alpha and beta particles, protons, and neutrons	Removes electrons and produces ions and reactive chemical species	DNA breaks, mutations, cell death, tissue injury, and cancer

Although UVA and UVB are usually classified as non-ionizing, they remain biologically harmful. UVA efficiently generates reactive oxygen species, whereas UVB can produce direct photochemical DNA lesions. These processes oxidize bases such as guanine and form thymine dimers and other cross-links. Such structural distortions can interfere with DNA replication and transcription.

3 Radioactive Decay and Radiation Measurement

The lecture reviewed several nuclear transformations:

- Alpha decay releases a helium nucleus.
- Beta-minus decay releases an electron and converts a neutron into a proton.
- Positron emission releases a positively charged electron.
- Electron capture occurs when a nucleus absorbs an orbital electron.
- Gamma decay releases electromagnetic energy without changing the element's atomic number.

Different units describe different aspects of radiation:

Q2. DNA Structure and Rectification Process

Note: This document interprets “rectification” as DNA repair. DNA replication is a separate process.

4 Structure of DNA

DNA, or deoxyribonucleic acid, is composed of two antiparallel polynucleotide strands twisted into a double helix.

Each DNA nucleotide contains:

- a deoxyribose sugar;
- a phosphate group; and
- one nitrogenous base: adenine (A), thymine (T), guanine (G), or cytosine (C).

The sugar and phosphate groups form the outer backbone of each strand. The nitrogenous bases face inward and pair through hydrogen bonds:

- adenine pairs with thymine through two hydrogen bonds; and
- guanine pairs with cytosine through three hydrogen bonds.

The two strands run in opposite directions. One strand runs from $5' \rightarrow 3'$, while the other runs from $3' \rightarrow 5'$. Complementary base pairing allows each strand to serve as a template during DNA replication and repair.

Within eukaryotic cells, DNA is wrapped around histone proteins to form nucleosomes. Nucleosomes are organized into chromatin, which can be further condensed into chromosomes.

5 DNA Repair Process

DNA is continually damaged by replication errors, ultraviolet light, ionizing radiation, reactive oxygen species, and chemical agents. Repair generally involves three stages:

1. The cell recognizes the damaged or incorrectly paired DNA.
2. The damaged base or section of DNA is removed or corrected.
3. DNA polymerase restores the correct sequence, and DNA ligase seals the sugar–phosphate backbone.

5.1 Direct Reversal

Direct reversal repairs certain chemical modifications without removing the affected nucleotide. For example, the enzyme O^6 -methylguanine-DNA methyltransferase (MGMT) can remove

particular alkyl groups from guanine.

5.2 Base-Excision Repair

Base-excision repair corrects small, chemically altered bases:

1. A DNA glycosylase recognizes and removes the damaged base.
2. An endonuclease cuts the DNA backbone at the resulting site.
3. DNA polymerase inserts the correct nucleotide.
4. DNA ligase seals the repaired strand.

5.3 Nucleotide-Excision Repair

Nucleotide-excision repair removes bulky lesions that distort the DNA double helix, including ultraviolet-induced pyrimidine dimers. The cell excises a short section of the damaged strand, DNA polymerase synthesizes a replacement section, and DNA ligase seals the remaining break.

5.4 Mismatch Repair

Mismatch repair corrects incorrectly paired bases and small insertion or deletion errors that escape proofreading during DNA replication. The repair system identifies the newly synthesized strand, removes the section containing the error, and replaces it with the correct sequence.

5.5 Single-Strand-Break Repair

When only one DNA strand is broken, repair proteins process the damaged ends, DNA polymerase replaces any missing nucleotides, and DNA ligase reconnects the strand. The intact complementary strand helps preserve the correct genetic information.

5.6 Double-Strand-Break Repair

Double-strand breaks are especially dangerous because both DNA backbones are severed. Cells use two principal repair pathways:

- **Homologous recombination** uses an undamaged homologous DNA sequence, usually the sister chromatid, as a template. This process is generally highly accurate.

- **Non-homologous end joining** reconnects the broken ends directly. It is faster and does not require a homologous template, but it may introduce small mutations at the repair site.

6 Cellular Response to Severe Damage

Cell-cycle checkpoints temporarily stop cell division while repair occurs. If the damage cannot be repaired, the cell may enter permanent growth arrest or undergo programmed cell death, known as apoptosis. If defective repair allows a damaged cell to survive and continue dividing, mutations may accumulate and contribute to cancer.

In radiation biophysics, double-strand breaks and clustered lesions are particularly important because they are more difficult to repair accurately than isolated base damage or single-strand breaks.

7 Conclusion

DNA's complementary double-stranded structure provides both stable storage of genetic information and a template for error correction. Multiple repair pathways protect this information by recognizing different forms of damage, restoring the correct nucleotide sequence, and resealing the DNA backbone. The success or failure of these pathways strongly influences mutation, cellular survival, aging, and disease development.